Process & Decision Documentation

Project/Assignment Decisions

Side Quests and A4 (Individual Work)

Edited Minecraft Side Quest 6 to include a debugging screen with controls for testing game mechanics against the environment. I modified the dashboard to include pressing ‘1’ for Boar Probes, ‘2’ for Colliders, ‘3’ for Invincible, and ‘4’ for Win Score = 1. Specified values change upon input from OFF to ON in the Minecraft platform game.

Default game:

Modified game with debugging:

Code Changes

Added debug flags

5 src/DebugOverlay.js

```diff
@@ -21,6 +21,9 @@ export class DebugOverlay {
 21  this.enabled = false;
 22  this.lines = [];
 23  this.maxLines = 8;
+24 +
+25 +  this.moonGravity = false;
+26 +  this.invincible = false;
 24  }
 25
 26  toggle() {
@@ -70,4 +73,4 @@ export class DebugOverlay {
 70
 71  camera.on();
 72  }
-73 - }
+76 + }
```

Toggle moon gravity with G

5 src/DebugOverlay.js

```diff
@@ -43,6 +43,11 @@ export class DebugOverlay {
 43
 44  camera.off();
 45
+46 +  if (kb.presses("g")) {
+47 +    this.moonGravity = !this.moonGravity;
+48 +    world.gravity.y = this.moonGravity ? 2 : 10;
+49 +  }
+50 +
 46  push();
 47  noStroke();
 48  fill(0, 160);
```

Toggle invincible with I

4 src/DebugOverlay.js

```diff
@@ -48,6 +48,10 @@ export class DebugOverlay {
 48  world.gravity.y = this.moonGravity ? 2 : 10;
 49  }
 50
+51 +  if (kb.presses("i")) {
+52 +    this.invincible = !this.invincible;
+53 +  }
+54 +
 51  push();
 52  noStroke();
 53  fill(0, 160);
```

Applying invincible every frame

4 src/DebugOverlay.js

```diff
@@ -52,6 +52,10 @@ export class DebugOverlay {
 52  this.invincible = !this.invincible;
 53  }
 54
+55 +  if (this.invincible && game?.level?.playerCtrl?.player) {
+56 +    game.level.playerCtrl.player.invulnTimer = 9999;
+57 +  }
+58 +
 55  push();
 56  noStroke();
 57  fill(0, 160);
```

Return to default

16 src/DebugOverlay.js

```diff
@@ -21,9 +21,6 @@ export class DebugOverlay {
     this.enabled = false;
     this.lines = [];
     this.maxLines = 8;
-
-    this.moonGravity = false;
-    this.invincible = false;
   }

   toggle() {
@@ -43,19 +40,6 @@ export class DebugOverlay {

     camera.off();

-    if (kb.presses("g")) {
-      this.moonGravity = !this.moonGravity;
-      world.gravity.y = this.moonGravity ? 2 : 10;
-    }
-
-    if (kb.presses("i")) {
-      this.invincible = !this.invincible;
-    }
-
-    if (this.invincible && game?.level?.playerCtrl?.player) {
-      game.level.playerCtrl.player.invulnTimer = 9999;
-    }
-
     push();
     noStroke();
     fill(0, 160);
```

Changed toggle and log

19 src/DebugOverlay.js

```diff
@@ -18,22 +18,15 @@

 export class DebugOverlay {
   constructor() {
-    this.enabled = false;
-    this.lines = [];
-    this.maxLines = 8;
+    this.probes = false;
+    this.colliders = false;
+    this.invincible = false;
+    this.easyWin = false;
   }

-  toggle() {
-    this.enabled = !this.enabled;
-  }
+  toggle() {}

-  log(evt) {
-    // evt is { name, payload } from EventBus wildcard
-    if (!evt) return;
-    const msg = `${evt.name}`;
-    this.lines.unshift(msg);
-    if (this.lines.length > this.maxLines) this.lines.length = this.maxLines;
-  }
+  log() {}

   draw({ game } = {}) {
     if (!this.enabled) return;
```

Created a onOff function

7 src/DebugOverlay.js

```diff
@@ -28,6 +28,13 @@ export class DebugOverlay {

   log() {}

+  onOff(val) {
+    if (val) {
+      return "ON";
+    }
+    return "OFF";
+  }
+
   draw({ game } = {}) {
     if (!this.enabled) return;
```

Created key triggers

13 src/DebugOverlay.js

```diff
@@ -36,7 +36,18 @@ export class DebugOverlay {
   }

   draw({ game } = {}) {
-    if (!this.enabled) return;
+    if (kb.presses("1")) {
+      this.probes = !this.probes;
+    }
+    if (kb.presses("2")) {
+      this.colliders = !this.colliders;
+    }
+    if (kb.presses("3")) {
+      this.invincible = !this.invincible;
+    }
+    if (kb.presses("4")) {
+      this.easyWin = !this.easyWin;
+    }

     camera.off();
```

Updated win condition

16 src/DebugOverlay.js

```diff
@@ -29,10 +29,11 @@ export class DebugOverlay {
  log() {}

  onOff(val) {
-   if (val) {
     return "ON";
   }
-  return "OFF";
 }

 draw({ game } = {}) {
@@ -49,6 +50,17 @@ export class DebugOverlay {
     this.easyWin = !this.easyWin;
   }

+  let lvl = game.level;
+  allSprites.debug = this.colliders;
+  if (this.invincible) {
+    lvl.player.invulnTimer = 9999;
+  }
+  if (this.easyWin) {
+    lvl.WIN_SCORE = 1;
+  } else {
+    lvl.WIN_SCORE = 15;
+  }
+
   camera.off();

   push();
```

Created debugging display

35 src/DebugOverlay.js

```diff
@@ -25,7 +25,6 @@ export class DebugOverlay {
  }

 toggle() {}
-
 log() {}

 onOff(val) {
@@ -62,35 +61,15 @@ export class DebugOverlay {
   }

   camera.off();
-  push();
   noStroke();
-  fill(0, 160);
-  rect(6, 6, 228, 86, 6);
-  pop();
-
   fill(255);
-  textSize(10);
-
-  const lvl = game?.level || null;
-  const score = lvl?.score ?? 0;
-
-  const playerCtrl = lvl?.playerCtrl || null;
-  const hp = playerCtrl?.health ?? "?";
-  const maxHp = playerCtrl?.maxHealth ?? "?";
-  const dead = playerCtrl?.dead ?? false;
-  const won = lvl?.won ?? false;
-
-  text(`Score: ${score}`, 12, 22);
-  text(`Health: ${hp}/${maxHp}`, 12, 34);
-  text(`won: ${won}  Dead: ${dead}`, 12, 46);
-
-  let y = 62;
-  for (const line of this.lines) {
-    text(line, 12, y);
-    y += 10;
-  }
-
   camera.on();
 }
}
```

Updated debug display

12 src/DebugOverlay.js

```diff
@@ -63,13 +63,13 @@ export class DebugOverlay {
   camera.off();
   fill(0, 150);
   noStroke();
-  rect(5, 5, 130, 55);
   fill(255);
-  textSize(8);
-  text("1: Boar Probes " + this.onOff(this.probes), 10, 17);
-  text("2: Colliders " + this.onOff(this.colliders), 10, 27);
-  text("3: Invincible " + this.onOff(this.invincible), 10, 37);
-  text("4: Win Score=1 " + this.onOff(this.easyWin), 10, 47);
+  rect(5, 5, 130 + 65, 55);
+  textSize(12);
+  text("Press '1' Boar Probes: " + this.onOff(this.probes), 10, 17);
+  text("Press '2' Colliders: " + this.onOff(this.colliders), 10, 27);
+  text("Press '3' Invincible: " + this.onOff(this.invincible), 10, 37);
+  text("Press '4' Win Score = 1: " + this.onOff(this.easyWin), 10, 47);
   camera.on();
 }
}
```

Updated debugging screen location

10 src/DebugOverlay.js

```diff
@@ -63,13 +63,13 @@ export class DebugOverlay {
63      camera.off();
64      fill(0, 150);
65      noStroke();
66  -   rect(5, 5, 130 + 65, 55);
67      fill(255);
68      textSize(12);
69  -   text("Press '1' Boar Probes: " + this.onOff(this.probes), 10, 17);
70  -   text("Press '2' Colliders: " + this.onOff(this.colliders), 10, 27);
71  -   text("Press '3' Invincible: " + this.onOff(this.invincible), 10, 37);
72  -   text("Press '4' Win Score = 1: " + this.onOff(this.easyWin), 10, 47);
73      camera.on();
74    }
75  }
```

```diff
63      camera.off();
64      fill(0, 150);
65      noStroke();
66  +   rect(5, 20, 130 + 65, 50);
67      fill(255);
68      textSize(12);
69  +   text("Press '1' Boar Probes: " + this.onOff(this.probes), 10, 17 + 15);
70  +   text("Press '2' Colliders: " + this.onOff(this.colliders), 10, 27 + 15);
71  +   text("Press '3' Invincible: " + this.onOff(this.invincible), 10, 37 + 15);
72  +   text("Press '4' Win Score = 1: " + this.onOff(this.easyWin), 10, 47 + 15);
73      camera.on();
74    }
75  }
```

GenAI Documentation

No GenAI was used.